

ARJUN PRASAD

Senior Backend Engineer | Distributed Systems & Payments

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SUMMARY

Backend engineer with 4.5+ years building high-throughput payment systems at PhonePe (500M+ users). Currently on the UPI Core Transactional team handling 20,000+ RPS and 500M+ daily transactions. Previously owned BBPS infrastructure processing 4M+ transactions/day. Built 2 greenfield systems, created a platform framework cutting infra costs 45%, led Active-Active DC migration with circular replication, and overhauled a pipeline reducing stuck transactions from 20K/day to <100. Looking for Senior Backend Engineer roles.

SKILLS

Java (8-17) | Dropwizard, Spring Boot, Hibernate, JUnit | MariaDB (Sharded), Aerospike | RabbitMQ, MemQ, Event-driven Architecture
Microservices, Active-Active DC, Docker, CI/CD, Nginx | System Design, Distributed Systems, DB Sharding, REST APIs, DR/BCP

EXPERIENCE

PhonePe -- UPI Core Transactional Team

Mar 2026 - Present

SDE-2 (Backend) | 20,000+ RPS | 500M+ UPI transactions/day | Pune

- Operating on India's highest-traffic payment rail -- 20K+ RPS peak, mission-critical availability
- Building E2C refund status tracking for 2-2.5 lakh daily refunds where users currently have zero in-app visibility
- Owning NPCI contract architecture, UDIR (UPI Dispute Resolution) flows, and external PSP integrations (GPay to PhonePe)

PhonePe -- BBPS Pod

Feb 2022 - Feb 2026

Software Engineer to SDE-2 (Backend) | 500 RPS peak | 4M+ billpay transactions/day | Pune

UPMS -- Greenfield Microservice (0 to 1)

- Designed and shipped a new microservice independently -- sharded DB, proxy layer, recon, biller kill-switches -- zero to 500+ RPS in 5 months
- Replaced polling with push-based presentment driving 20% transaction growth; completed NPCI certification and monolith migration

Axon -- Async Messaging Framework

- Created reusable framework abstracting RMQ/MemQ with unified publish API, fallback strategies, dynamic worker buckets -- adopted org-wide
- 45% infrastructure cost reduction across PhonePe; per-service integration time reduced from days to minutes

Active-Active Data Center Migration

- Redesigned DB schema for circular replication across DCs, migrated async workers to MemQ -- zero-downtime failover, 50% blast radius reduction

Status Check Pipeline Overhaul

- Reduced stuck transactions from 20,000+/day to <100; resolution time from 9+ hours to <10 min; eliminated wrong refund risk

Other Highlights

- Piped Gas Meter Reading: built contract-to-production in 1 week, demonstrated at Global Fintech Festival 2025 (PhonePe-exclusive)
- Paylink migration: +15% monthly transactions, zero downtime | Loan API 1.1: led NPCI spec compliance independently
- Launched 6 bill categories; created SOP eliminating eng dependency | Mentoring 3 engineers | 117 code reviews | 6 DR drills

Samsung Research Institute

Jul 2021 - Feb 2022

Software Engineer | Noida

- Reduced audio device setup time by 30% through Bluetooth module optimization (1,000+ devices)
- Samsung SWC Professional certification (Data Structures & Algorithms)

EDUCATION

B.Tech in Computer Science & Engineering

2017 - 2021

National Institute of Technology, Delhi

RECOGNITION

- "Distinguished" rating (highest tier) at PhonePe -- "Completed the implementation flawlessly and quickly" -- Engineering Manager
- Scale progression: built service at 500 RPS, now operating at 20,000+ RPS -- consistently entrusted with increasing scope and ownership
- 345+ merge requests | 605 tasks delivered | 356 architecture docs | 4 consecutive strong performance cycles | GFF 2025 demo